KIIT Deemed to be University
Online End Semester Examination(Spring Semester-2021)

Relational Database Management System – CS3044

Applicable to Courses: 6th Semester, B.Tech

Full Marks=50 **Time:2 Hours**

SECTION-A (Answer All Questions. Each question carries 2 Marks)

Time:30 Minutes **(7×2=14 Marks)**

Question No	Question Type (MCQ/ SAT)	Question	CO Mapping	Answer Key (For MCQ Questions only)
Q.No:1	MCQ	Which one of the following is used to define the structure of the relation, deleting relations and relating schemas? a) DML(Data Manipulation Langauge) b) DDL(Data Definition Langauge) c) Query d) Relational Schema	CO 1	d) DDL (Data Definition Langauge)
	MCQ	The database schema is written in _____? a) HLL b) DML c) DDL d) DCL	CO1	c) DDL
	MCQ	Data integrity constraints are used to: a) Control who is allowed access to	CO1	c) Improve the quality of data

		the data b) Ensure that duplicate records are not entered into the table c) Improve the quality of data entered for a specific property (i.e., table column) d) Prevent users from changing the values stored in the table		entered for a specific property (i.e., table column)
	MCQ	What is an Instance of a Database? a) The logical design of the database system b) The entire set of attributes of the Database put together in a single relation c) The state of the database system at any given point of time d) The initial values inserted into the Database immediately after its creation	CO1	c) The state of the database system at any given point of time
Q.No:2	**MCQ**	Which symbol represents weak entity sets in ER- model? a) Dotted rectangle b) Diamond c) Dotted outlined rectangle d) Dotted Diamond	CO2	c) Dotted outlined rectangle
	MCQ	How do we convert a weak entity set to a strong entity set? a) By applying generalization b) By applying aggregation c) By adding appropriate attributes d) Cannot be converted	CO2	c)By adding appropriate attributes
	MCQ	An attribute that make more than one single attributes is called ____? a) Multi valued attribute b) Derived attribute c) Composite attribute d) Single value attribute	CO2	a) Multi valued attribute
	MCQ	Entities which are associated with other entities via a relationship set is called as ?	CO2	b) Cardinality

		a) Schema b) Cardinality c) Attribute d) Entity relationship		
Q.No:3	**MCQ**	Which forms simplifies and ensures that there are minimal data aggregates and repetitive groups: a) 1NF b) 2NF c) 3NF d) BCNF	CO4	c) 3NF
	MCQ	A table in which normal for removes all the hidden dependencies? a) 4NF b) 2NF c) 3NF d) BCNF	CO4	b) 2NF
	MCQ	Which one of the following statements about normal forms is FALSE? a) Lossless, dependency -preserving decomposition into BCNF is always possible b) Lossless, dependency -preserving decomposition into 3NF is always possible c) BCNF is stricter than 3NF d) Any relation with two attributes is in BCNF	CO4	a) Lossless, dependency -preserving decompositi on into BCNF is always possible
	MCQ	Which one of the following statements if FALSE? a) Any relation with two attributes is in BCNF b) A relation in which every key has only one attribute is in 2NF c) A prime attribute can be transitively dependent on a key in a BCNF relation. d) A prime attribute can be transitively dependent on a key in a 3 NF relation.	CO4	c) A prime attribute can be transitively dependent on a key in a 3 NF relation.
Q.No:4	**MCQ**	Consider a schema R(A, B, C, D) and functional dependencies A -> B and C -> D. Then the decomposition of R into R1 (A, B) and R2(C, D) is a) Dependency preserving and lossless join b) Lossless join but not	CO4	c) Dependency preserving but not lossless join

		dependency preserving c) Dependency preserving but not lossless join d) Not dependency preserving and not lossless join		
	MCQ	Consider a schema R(A, B, C) and functional dependencies A -> B and B -> C. Then the decomposition of R into R1 (A, B) and R2(B, C) is a) Dependency preserving and lossless join b) Lossless join but not dependency preserving c) Dependency preserving but not lossless join d) Not dependency preserving and not lossless join	CO4	a) Dependency preserving and lossless join
	MCQ	Consider the following functional dependencies in a database. Date_of_Birth->Age Age->Eligibility Name->Roll_number Roll_number->Name Course_number->Course_name Course_number->Instructor (Roll_number, Course_number)->Grade The relation (Roll_number, Name, Date_of_birth, Age) is: a) in second normal form but not in third normal form b) in third normal form but not in BCNF c) in BCNF d) in none of the above	CO4	e) In none of the above
	MCQ	The relation schema Student_Performance (name, courseNo, rollNo, grade) has the following FDs: name,courseNo->grade rollNo,courseNo->grade name->rollNo rollNo->name	CO4	b) 3NF

		The highest normal form of this relation scheme is a) 2NF b) 3NF c) BCNF d) 4NF		
Q.No:5	**MCQ**	Statement 1: A tuple is a row in a relation Statement 2: Existence of multiple foreign keys in a same relation is possible a) Both the statements are true b) Statement 1 is correct but Statement 2 is false c) Statement 1 is false but Statement 2 is correct d) Both the statements are false	CO1, CO5	a) Both statements are true
	MCQ	Data integrity constraints are used to: a) Control who is allowed access to the data b) Ensure that duplicate records are not entered into the table c) Improve the quality of data entered for a specific property (i.e., table column) d) Prevent users from changing the values stored in the table	CO1, CO2	c) Improve the quality of data entered for a specific property (i.e., table column)
	MCQ	Choose the correct statement regarding superkeys a) A superkey is an attribute or a group of multiple attributes that can uniquely identify a tuple b) A superkey is a tuple or a set of multiple tuples that can uniquely identify an attribute c) Every superkey is a candidate key d) A superkey is an attribute or a set of attributes that distinguish the relation from other relations	CO2, CO5	a) A superkey is an attribute or a group of multiple attributes that can uniquely identify a tuple
	MCQ	Which one of the following given statements possibly contains the error? a) select * from emp where empid	CO5	d) select empid where empid is = 10003 and Lname = 'L%'

		= 10003; b) select empname from emp where empid = '0003; c) select * from emp where empname = 'L%' d) select empid where empid is = 10003 and Lname = 'L%'		
Q.No:6	**MCQ**	What do you mean by One to many relationship? a) one student may have many teachers b) one teacher can have many students c) many students can have 1 teacher d) many teachers can have 1 student	CO1, CO2	a) one student may have many teachers
	MCQ	The _____ is the one in which the primary key of one relation is used as a normal attribute in another relation. a) Referential relation b) Referencing relation c) Referenced relation d) Referred relation	CO5	c) Referenced relation

	MCQ	To change the datatype of a column which is the correct statement?
		
		a) ALTER TABLE EMP add (Address Char(24), Dob date, DeptNo number(8));
		
		b) ALTER TABLE EMP add (Address Char(24)), add(Dob date), add (DeptNo number(8));
		
		c) ALTER TABLE EMP add (Address Char(24), Dob(date), add (DeptNo number(8));
		
		d) ALTER TABLE EMP add (Address Char(24) Dob(date) (DeptNo number(8));
			CO5	a) ALTER TABLE EMP add (Address Char(24),Dob date,DeptNo number(8));

	MCQ	To change the datatype of a column which is the correct statement a) ALTER TABLE table_name update column_name	CO5	c) ALTER TABLE table_name MODIFY column_name

		column_type; b) ALTER TABLE table_name rename column_name column_type; c) ALTER TABLE table_name MODIFY column_name column_type; d) ALTER TABLE table_name rechange column_name column_type;		column_type;
Q.No:7	**MCQ**	Which of the following is a top-down approach in which the entity's higher level can be divided into two lower sub-entities? a) Aggregation b) Generalization c) Specialization d) All of the above	CO3	c) Specialization
	MCQ	Which of the following is used in the application programs to request data from the database management system? a) Data Definition Language b) Data Manipulation Language c) Data Control Language d) All of the above	CO5	b) Data Manipulation Language
	MCQ	The database management system can be considered as the collection of _____ that enables us to create and maintain the database. a) Translators b) key c) Programs d) Language Activity	CO1	c) Programs
	MCQ	Which of the following SQL statements is correct? a) TRUNCATE * FROM TABLE Sales b) TRUNCATE TABLE Sales c) TRUNCATE Sales TABLE d) None of the above	CO5	b) Truncate table Sales

SECTION-B(Answer Any Three Questions. Each Question carries 12 Marks)

Time: 1 Hour and 30 Minutes **(3×12=36 Marks)**

Question No	Question	CO Mapping (Each question should be from the same CO(s))
Q.No:8	1. What were the problems with traditional file systems and what approaches were taken to overcome the disadvantages of traditional file systems? 2. Discuss the current practices for managing/abstracting data from the database systems along with its advantages.	CO1, CO3, CO4
	1. 3-level data abstraction is an important consideration for any database management system-justify your answer with proper explanation and suitable example. 2. How the three-schema architecture separate the user applications and physical database. 3. Why do we need three-schema architecture? Also, illustrate the difference between logical data independence and physical data independence.	
	1. What are the database design issues and challenges? 2. What are the problems caused by Redundancy? 3. Explain about Normalization and need for normalization. Explain about Third NF and BCNF with relevant table structure.	
Q.No:9	1. Explain the use of ER model. How the ER model present basic object and their properties. • A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c)	CO1, CO2

	students, including student-id, name, and program; and (d) instructors, including identification number, name, department, and title. Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints. Also convert the ER diagram into relational schemas. 2. We can convert any weak entity set to a strong entity set by simply adding appropriate attributes. Why, then, do we have weak entity sets?	
	1. What is the purpose of designing a conceptual model before the physical model? 2. Consider a database used to record the marks that students get in different exams of different course offerings. 3. Construct an E-R diagram that uses only a binary relationship between students and course-offerings. Make sure that only one relationship exists between a particular student and course-offering pair, yet you can represent the marks that a student gets in different exams of a course offering. Write the relational schema for the above ER-Diagram.	
	Draw the entity-relationship diagram for the different industries as below: There are many industries (identified by unique inid along with name, location and contact number(s)). Each industry has multiple departments. Each department has one unique did along with extension number. Each departments is producing multiple products. Each product has one unique prid along with product description, price and label. Many workers (identified by unique wid along with wname, salary, doj and address (can be decomposed to street, city and pin) as attributes) are working in the different departments; but, one worker can work in one department only. The workers are managing the different products. Make necessary assumptions. Also convert the ER diagram into relational schemas. Convert the ER diagram into Relational model	
Q.No:10	Write the Relational algebra query for the below schema and given query. Shipment Database Customer(cust_id,cust_name, annual_revenue)	CO_5, CO_1

	Truck(truckno, driver_name) City(city_name, population) Shipment(shipment_no, cust_id, weight, truckno, destination_city) i) Query: Find the names of drivers who have delivered shipments weighing over 100 pounds ii) iii) Query: List the cities which have received shipments from customers having over $15 million in annual revenue ii) Query: List the customers having over $5 million in annual revenue who have sent shipments weighing greater than 1 pound iv) Query: List the customers whose shipments have been delivered by truck driver Ramesh v) Query: Find the customers having over $5 million in annual revenue who have sent shipments weighing less than 1 pound or have sent a shipment to Bhubaneswar vi) : Find the drivers who have delivered shipments to every city vii) List the drivers who have delivered shipments for customers with annual revenue over $20 million to cities with population over 1 million	
	Write the Relational algebra query for the below schema and given query. Shipment Database Customer(cust_id,cust_name, annual_revenue) Truck(truckno, driver_name) City(city_name, population) Shipment(shipment_no, cust_id, weight, truckno, destination_city) i) Query: Find the names of customers with more than $10 million in annual revenue ii) Query: Find the driver of truck 45 iii) Query: Find the name and annual revenue of customers who have sent shipments weighing over 100 pounds iv) Query: Find the names of drivers who have delivered shipments weighing over 100 pounds v) Query: Find the customers who have sent shipments to every city with population over 500000 vi) Query: List the customers whose shipments have been delivered by truck driver Ramesh	

	Write the Relational algebra query for the below schema and query. Shipment Database Customer(cust_id,cust_name, annual_revenue) Truck(truckno, driver_name) City(city_name, population) Shipment(shipment_no, cust_id, weight, truckno, destination_city) i) Query: Find the truck numbers of trucks which have carried shipments weighing over 100 pound ii) Query: List the customers having over $5 million in annual revenue who have sent shipments weighing greater than 1 pound iii) Query: Find the drivers who have delivered shipments to every city iv) Query: Find the name and annual revenue of customers who have sent shipments weighing over 100 pounds v) Find the names of cities which have received shipments weighing over 100 pounds vi) Query: Find the customers having over $5 million in annual revenue who have sent shipments weighing less than 1 pound or have sent a shipment to Bhubaneswar	
Q.No:11 pro	What is the need of normalizing a database? What is the minimal normal form that a relation must satisfy? Explain with examples. Examine the table below: **staffNo** | **branchNo** | **branchAddress** | **name** | **position** | **hoursPerWeek** S0001 | B01 | Damana, BBSR | Ramesh | Assistant | 16 S0002 | B02 | Patia, BBSR | Prayas | Assistant | 12 S0001 | B02 | Damana, BBSR | Ramesh | Assistant | 8 S0002 | B01 | Patia, BBSR | Prayas | Assistant | 12 Functional dependencies are:	CO4, CO6

	branchNo -> branchAddress staffNo -> name, position staffNo, branchNo -> hoursPerWeek (a) Why is this table not in 2NF? (b) Describe and illustrate the process of normalizing the data shown in this table to third normal form (3NF). (c) Identify the primary, (alternate) and foreign keys in your 3NF relations.	
	1. Relation R has eight attributes A, B, C, D, E, F, G and H. Fields of R contain only atomic values. F = {CH → G, A → BC, B → CFH, E → A, F → EG} is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have? 2. Differentiate between a serial schedule and a concurrent schedule with suitable examples.	
	1. Consider a relation R(A, B, C, D, E, F) and the set F={A → BC, E → CF, & CD → EF} Find the closure of {A,B}, List the prime attributes and non-prime attributes. With the given set of functional dependencies at present the above schema is in which normal form? 2. What are different types of anomalies? Explain with examples.	